

NOURHAN EL-SEHRAWY

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SUMMARY

Business-minded Data Scientist with four years of experience delivering production-ready solutions in fraud detection, recommendation systems, and BI automation. Proficient in Python, SQL, Power BI, and AWS, with current work focused on integrating LLMs and generative AI into internal tools. Experienced with MLOps practices, containerization (Docker), and cloud deployment (AWS SageMaker). Passionate about scalable analytics, real-time data pipelines, and building intelligent systems that drive measurable business outcomes.

WORK EXPERIENCE

Data Scientist @ Jumia Group. (Nov. 2023 - Present)

Started in Internal Audit; transitioned to Core Data Science team.

- Improved "View Together" recommendation accuracy by 15% by implementing and deploying an A/B test comparing collaborative filtering strategies based on product similarity; collaborated with the product team to validate uplift through CTR analysis.
- Reduced product miscategorization errors by 20% by applying modern embedding techniques (e.g., Word2Vec, BERT-style encodings) and unsupervised clustering to classify and correct mislabeled items.
- Deployed Power BI dashboards used weekly by compliance teams, improving internal audit visibility and efficiency.
- Standardized analytics workflows across platforms by collaborating with cross-functional engineering, audit, and product teams.

Skills: Python, SQL, Power BI, PySpark, Docker, AWS SageMaker, Recommender Systems, LLMs, Fraud Detection, Statistical Data Analysis, Exploratory Data Analysis, A/B testing.

Data Scientist @ Chefaa Egypt. (Nov. 2021- Nov. 2023)

- Built and deployed a Business Intelligence Hub as a web-based analytics platform (Flask + SQL + Power BI), enabling real-time insights for internal teams and partners.
- Generated over 500k EGP in revenue through the BI Hub, making it the company's highest-earning analytics product across all verticals.
- Automated internal reporting and partner analytics, reducing manual workload by 70% and accelerating access to performance metrics.
- Designed and maintained data pipelines to support dashboards for sales, logistics, and operations teams.
- Collaborated with business stakeholders to define KPIs and build tailored data solutions that informed strategic decisions.
- Conducted exploratory and statistical data analysis to uncover user behavior patterns and operational inefficiencies.

Skills: Python (Programming Language) · SQL · Statistical Data Analysis · Exploratory Data Analysis · Docker · Amazon Web Services (AWS)

Data Annotation & Machine Learning intern @ Master Linux. (July 2021- Sep 2021)

- Labeled and structured datasets for machine learning teams, supporting the development of NLP and computer vision models.
- Gained hands-on experience in deep learning, reinforcement learning, and NLP pipelines through close collaboration with ML engineers.
- Contributed to internal tools and learned industry-standard annotation workflows and evaluation practices.

Skills: Reinforcement Learning · Data Science · Python (Programming Language)

EDUCATION

Master of Science in Data Science

Faculty of Graduate Studies for Statistical Research, Cairo, *Cairo University* (2024 – present)

GPA: 3 (Very good)

Bachelor of Science in Computer Science

Computer & Artificial Intelligence, Cairo, *Helwan University* (2017 – 2021)

GPA: 3 (Very good)

PROJECTS

- Drug-Drug Interaction (Graduation Project – A+ Grade)** [see project](#)
Developed a machine learning system to predict harmful drug interactions using the DrugBank dataset.
 - Trained and evaluated multiple models (Random Forest, SVM, MLP) to classify drug pair interactions based on molecular and pharmacological data.
 - Achieved high classification accuracy, with MLP outperforming traditional models on precision and recall.
 - Hosted full code and documentation on GitHub for reproducibility and review.**Tools & Techniques:** Python, Scikit-learn, Pandas, Flask, Android Studio (for app), Model Evaluation, Classification Metrics

- **Vector Space Model** [see project](#)
Built a search engine prototype using the Vector Space Model (VSM) to retrieve relevant documents based on user queries.
 - Implemented a full-text search pipeline using TF-IDF weighting to rank document relevance.
 - Preprocessed input data using NLTK (tokenization, stop-word removal, lemmatization) for clean vectorization.
 - Deployed the application via Flask, allowing users to input custom queries and receive ranked search results in real time.
 - Demonstrated the fundamentals of information retrieval and NLP in a lightweight, interactive web interface.**Tools & Techniques:** Python, NLTK, Flask, TF-IDF, Cosine Similarity, Information Retrieval
- **Credit Card Fraud Detection** [see project](#)
Trained classification models to detect fraudulent transactions using imbalanced real-world data.
 - Applied techniques like SMOTE and undersampling to handle class imbalance.
 - Evaluated models using precision, recall, and AUC to optimize detection without excessive false positives.**Tools & Techniques:** Python, Scikit-learn, Pandas, SMOTE, ROC AUC
- **Analyze AB Test Results** [see project](#)
Analyzed conversion data from an e-commerce A/B test to assess the impact of a new landing page.
 - Applied statistical testing (z-test) to evaluate significance and reliability.
 - Delivered insights and recommendations to support product decision-making.**Tools & Techniques:** Python, Pandas, SciPy
- **Breast Cancer Prediction with R** [see project](#)
Built a classification model to predict malignancy based on diagnostic features.
 - Evaluated model performance using metrics like accuracy, precision, and recall.
 - Used R for data preprocessing, modeling, and visualization.**Tools & Techniques:** R, caret, ggplot2, Logistic Regression, Decision Trees
[check my github for other projects](#)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL (PostgreSQL, MySQL), Bash
- **Machine Learning & AI:** Scikit-learn, XGBoost, Recommender Systems, A/B Testing, Model Evaluation
- **Deep Learning & GenAI:** TensorFlow, Keras
- **NLP & Text Analytics:** NLTK, Regex, Text Preprocessing, Tokenization
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Streamlit, Excel
- **Data Engineering & Pipelines:** PySpark, SQL Pipelines, Feature Engineering, Airflow
- **Cloud & Deployment:** AWS (S3, SageMaker), Docker, Flask
- **Tools & Platforms:** Jupyter Notebook, VS Code, Git/GitHub, Postman, Dataiku, MLflow

SOFT SKILLS

- Self-motivated.
- Attention to details.
- Critical thinking.
- Communication.
- Team player.
- Story-telling.
- Curiosity.
- Leadership.
- Adaptability.

CERTIFICATES

- Certified Machine Learning Practitioner - Dataiku
- Data Analysis Professional Nanodegree Program – Udacity [see-credentials](#)
- Data Analysis Advanced Nanodegree Program – Udacity [see-credentials](#)
- Supervised Learning with scikit-learn – DataCamp [see-credentials](#)
- Problem Solving Campaigne [see-credentials](#)

VOLUNTEERING

Jan 2019 – Sep 2019	Photography & Video Production Member, <i>Minders</i>
Apr 2019 – Jun 2019	Public Relations Volunteer, <i>Minders</i>
Dec 2019 – Oct 2020	Photography & Video Production Head, <i>Minders</i>
Oct 2020 – July 2021	Technical Core Team member, <i>Google Student Developer Club - Helwan University</i>

LANGUAGES

- Arabic (Native)
- English (Advanced)
- German (Beginner)